

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I L. Dattu Kumar 192324141 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

L. Dattu Kumar

3. Concept Map - Classical Encryption Schemes

Classical Encryption Schemes

Two main Techniques :- 1. Substitution Technique
2. Transposition Technique.

1. Substitution Technique :-

* Replaces each plaintext symbol with another symbol.

* Data Structure :- The order of characters remains the same, but the characters are changed.

e.g. :- Caesar, Cipher, Monoalphabetic Cipher, Vignere Cipher, Hill Cipher.

Security :- Depends on the complexity of the key.

e.g. :- Plaintext :- HELLO

Caesar (+3) :- KHOO R

2. Transposition Techniques :-

* Rearranges the positions of plaintext characters.

* Data Structures :- Characters remain the same, but their order is changed.

e.g. :- Rail Fence Cipher, Columnar Transposition.

Security :- Depends on the complexity of the key.

e.g. :- Plaintext :- HELLO

Transposition :- LHOEL

Similarities :-

Substitution + Transposition → Classical Encryption.

- Uses a secret key/rule.
- Converts plaintext → ciphertext.
- Provides confidentiality.
- Can be attacked using cryptanalysis.

Differences :-

Feature	Substitution	Transposition
Basic Operation	Change symbols	Changes positions
Characters	Changed	Remain the same
Order	Mostly preserved	changed
Example	Caesar cipher	Rail Fence
Weakness	Frequency analysis	Pattern Analysis.

Data Structure & Security Level :-

Substitution → Changes the actual symbols.

- Frequency patterns may remain.
- Can be vulnerable to frequency analysis.

Transposition → Preserves the original symbols.

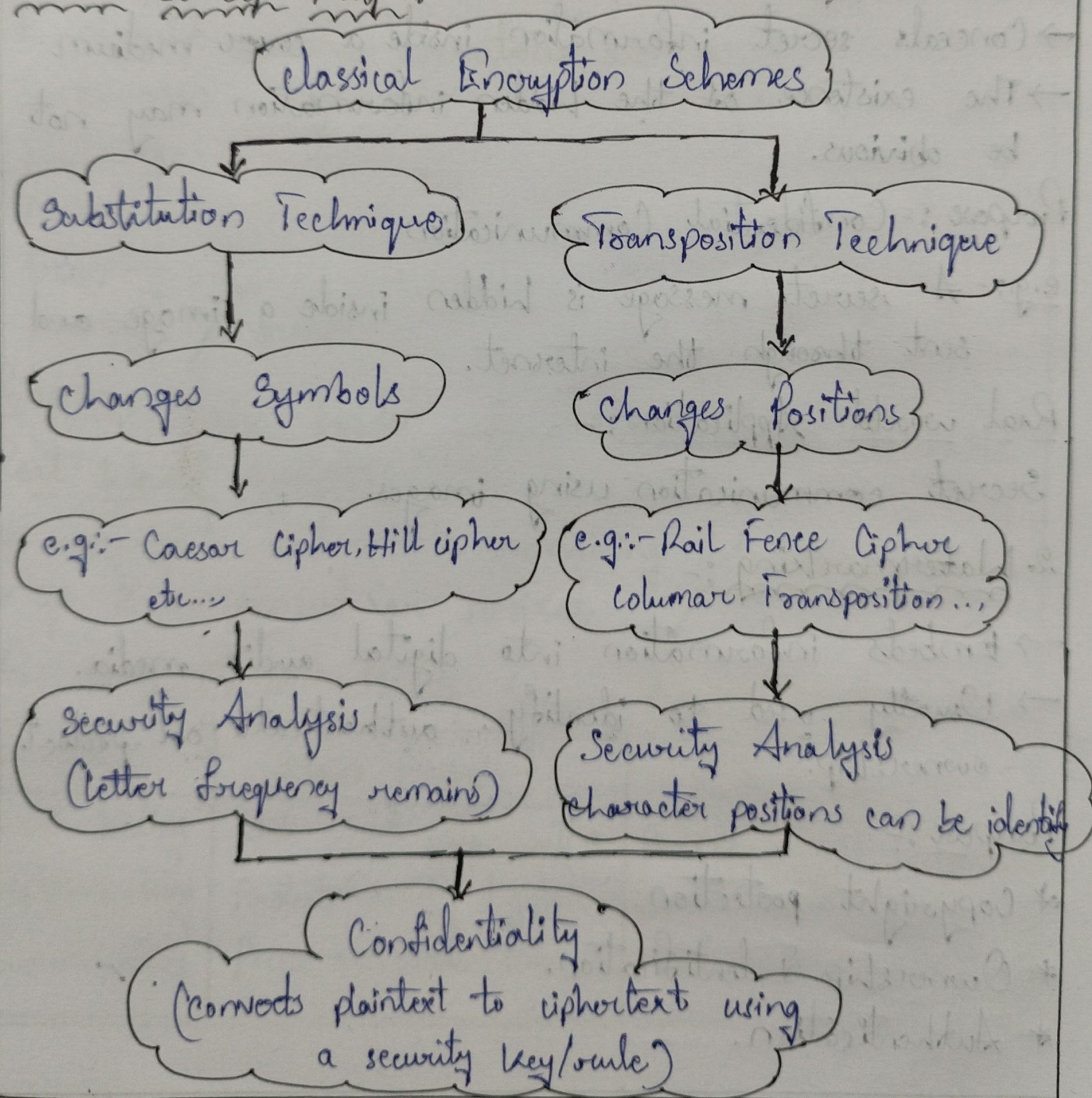
- Change their positions.
- Can reveal patterns when the key is weak

Which is more vulnerable?

Simple substitution techniques are generally vulnerable to frequency analysis, especially monoalphabetic ciphers, because the frequency of letters is preserved.

However the simple transposition is also vulnerable if the key or arrangement is weak.

Final Concept Map :-



4. Concept Map - Steganography

It hides information inside another medium.

Examples for cover media :-

* Image

* Audio

* Video

* Text

1. Data Hiding

→ Conceals secret information inside a cover medium

→ The existence of the hidden information may not be obvious.

Purpose :- Confidential Communication

e.g.:- A secret message is hidden inside a image and sent through the internet.

Real world Application :-

Secret communication using images.

2. Watermarking :-

→ Embeds information into digital audio media.

→ Usually used to identify, authenticate or protect ownership.

Purpose :-

* Copyright protection.

* Ownership Identification.

* Authentication.

e.g.: - A photographer adds a digital watermark to an image.

Real world Application:-
copyright protection of digital photographs.

3. Survivability:-

→ Measures how well the hidden information / watermark remains available after modifications or attacks.

It should survive operations such as:-

* Compression.

* Resizing.

* Noise.

* Cropping.

* Format Conversion.

Purpose:- Make hidden information remain detectable.

Real World Application:-

A copyright watermark remaining detectable after an image is compressed or resized.

How they Contribute to Secure Communication

Concept	Main Purpose	Contribution
Data Hiding	Hide secret information	Provide covert communication
Watermarking	Protect/identify digital	Provide ownership.
Survivability	Preserve hidden inform	Maintains security modify

Simple Real World Examples :-

Data Hiding → A secret message hidden inside a digital image

Watermarking → A company places its logo/watermark on an image to prevent unauthorized use.

Survivability → A copyright watermark remains detectable even after the image is compressed.

Final Answer In Concept Form

